

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701)

Topic 35 — Polymerisation

Paper 4 Topical Examination Pack & Comprehensive Examiner Mark Scheme

CANDIDATE	Urwah
SYLLABUS	Cambridge International A Level Chemistry (9701) — Paper 4
TOPIC COVERAGE	Condensation Polymers · Polyesters & Polyamides · Repeat Units & Monomers · Kevlar & Tensile Strength
COMPONENT	Paper 4 — A Level Structured Questions (Theory & Calculations)
TOTAL MARKS	220 Marks (50 Comprehensive Questions)

HOW TO USE THIS A LEVEL PAPER 4 PACK

1. Complete each structured question thoroughly in the answer spaces provided, adhering strictly to mark allocations.
2. Questions are classified into Core and Advanced tiers, covering multi-step chemical derivations, mechanisms, and numerical proofs.
3. Every question features authentic Cambridge A Level Paper 4 examination references (e.g. 9701/42/M/J/23) for benchmark cross-referencing.
4. Full point-by-point examiner mark schemes with step-by-step working and error-carried-forward (ECF) criteria are detailed at the end.

SYLLABUS SUBTOPICS COVERED

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|---|--|
| 35.1 Condensation Polymerisation Mechanisms | — Formation of polyesters (Terylene/PET) and polyamides (Nylon 6,6, Kevlar) with elimination of small molecules |
| 35.2 Deducing Monomers from Polymer Chains | — Deconstruction of complex copolymer chains; identifying ester and amide bonds; deducing constitutional isomers |
| 35.3 Kevlar & Advanced High-Performance Polymers | — Rigid rod structure of poly(p-phenylene terephthalamide); extensive regular intermolecular hydrogen bonding |
| 35.4 Polymer Degradability & Environmental Chemistry | — Hydrolytic susceptibility of ester and amide linkages in acidic and alkaline media; comparison with natural polymers |
| High-Frequency Core Repeats (Q41–Q50) | — The 10 most frequently tested Cambridge Paper 4 questions on Polymerisation from the past 10 years |

SECTION A: 6-MARK EXTENDED EXAM QUESTIONS (40% TARIFF · Q1–Q16)

1 Structures and Synthesis of Synthetic Condensation Polymers — 9701/42/M/J/23/Q11 [6 Marks]

HARD · Syllabus 35.1

The repeat units of two major synthetic polymers, Nylon 6,6 and Terylene, are shown in Fig. 1.1.

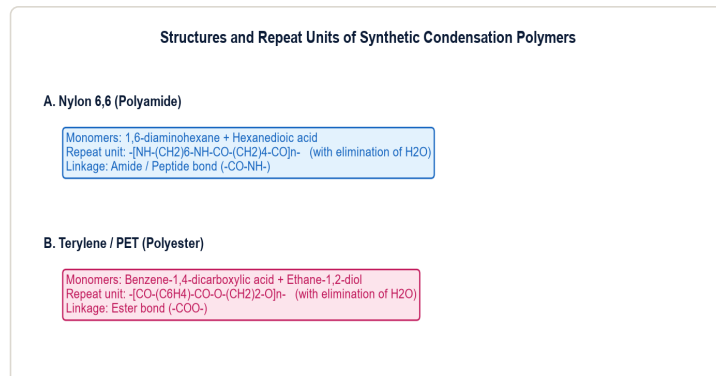


Fig. 1.1: Repeat units and linkage types of Nylon 6,6 (polyamide) and Terylene (polyester).

- (a) State the systematic IUPAC names of the two monomers required to synthesise Terylene. [2]

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- (b) Draw the repeat unit of Nylon 6,6 and clearly identify the functional linkage present. [2]

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- (c) Contrast addition polymerisation and condensation polymerisation in terms of monomer functional groups and reaction byproducts. [2]

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2 Chemical Degradability of Addition vs Condensation Polymers — 9701/41/O/N/22/Q9 [6 Marks]

HARD · Syllabus 35.4

The environmental persistence and chemical degradability of polymers depend heavily on their backbone structure as illustrated in Fig. 2.1.

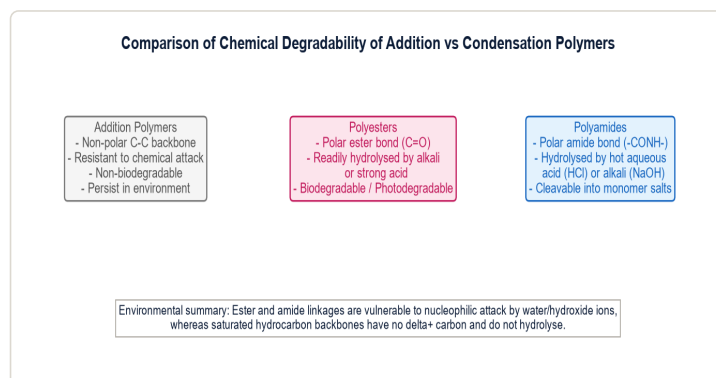


Fig. 2.1: Comparison of chemical hydrolytic resistance between polyalkenes, polyesters, and polyamides.

- (a) Explain why polyalkenes, such as poly(ethene), are non-biodegradable and persist in the environment for centuries. [2]

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- (b) Explain why polyesters and polyamides can be broken down in landfills, stating the mechanism of degradation. [2]
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- (c) Describe what happens chemically when Nylon 6,6 is treated with hot concentrated sodium hydroxide. [2]
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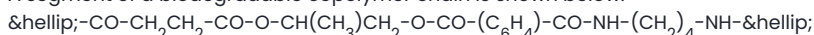
3 Structure, Intermolecular Forces, and Strength of Kevlar — 9701/42/M/J/22/Q10 [6 Marks] HARD · Syllabus 35.3

Kevlar is a high-strength aromatic polyamide used in bullet-resistant vests and aerospace composites.

- (a) Draw the structural formula of the repeat unit of Kevlar. [2]
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- (b) Name the two monomers used in the manufacture of Kevlar. [2]
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- (c) Explain in terms of molecular geometry and intermolecular bonding why Kevlar has an exceptionally high tensile strength. [2]
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4 Deducing Monomers from Complex Condensation Polymers — 9701/41/M/J/21/Q11 [6 Marks] HARD · Syllabus 35.2

A segment of a biodegradable copolymer chain is shown below:



- (a) Identify all types of linkages present along this polymer backbone. [2]
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- (b) Draw the structural formulas of the four distinct monomers that react together to form this copolymer. [4]
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5 Polylactic Acid (PLA): Synthesis from Lactic Acid and Biodegradability — 9701/42/O/N/20/Q9 [6 Marks] HARD · Syllabus 35.1

Lactic acid (2-hydroxypropanoic acid), $\text{CH}_3\text{CH(OH)COOH}$, can undergo self-condensation to form the biodegradable polyester poly(lactic acid), PLA.

- (a) Draw the repeat unit of poly(lactic acid). [2]
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- (b) Explain why lactic acid can form a polymer on its own, whereas ethanoic acid cannot. [2]
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- (c) State two environmental advantages of using PLA instead of poly(propene) for food packaging. [2]
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6 Synthesis of Nylon 6,6: Laboratory Interfacial Polymerisation — 9701/43/M/J/23/Q10 [6 Marks]

HARD · Syllabus 35.1

In the school laboratory demonstration known as the 'Nylon rope trick', two immiscible solutions are placed in a beaker:

- Solution 1: Hexane-1,6-diamine dissolved in aqueous sodium hydroxide
- Solution 2: Hexanedioyl dichloride dissolved in cyclohexane

- (a) Explain why polymerisation occurs only at the interface between the two liquid layers. [2]
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- (b) State the role of sodium hydroxide in Solution 1. [2]
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- (c) Explain why hexanedioyl dichloride is used in this demonstration rather than hexanedioic acid. [2]
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7 Photodegradable and Soluble Polymers: Norrish Reactions — 9701/42/F/M/22/Q11 [6 Marks]

HARD · Syllabus 35.4

- (a) Explain how the incorporation of carbonyl groups ($C=O$) into a polyalkene backbone makes the polymer photodegradable. [3]
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- (b) Poly(ethenol) is a synthetic polymer that is readily soluble in water. Draw the repeat unit of poly(ethenol) and explain its water solubility. [3]
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8 Natural Polymers: Proteins and Polysaccharides as Condensation Polymers — 9701/41/O/N/23/Q10 [6 Marks] **HARD · Syllabus 35.1**

- (a) State the type of reaction by which starch and cellulose are formed from glucose, and identify the small molecule eliminated. **[2]**

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- (b) Identify the specific linkage connecting glucose units in polysaccharides. **[1]**

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- (c) Compare the enzymatic hydrolysis of starch in the human body with the chemical hydrolysis of synthetic polyesters in industry. **[3]**

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9 Conducting Polymers: Poly(ethyne) and Delocalised π Systems — 9701/42/M/J/20/Q9 [6 Marks] **HARD · Syllabus 35.3**

- (a) Draw the skeletal formula of three repeat units of poly(ethyne), showing alternating single and double bonds. **[2]**

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- (b) Explain how the alternating conjugated double bond system enables poly(ethyne) to conduct electricity when 'doped'. **[3]**

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- (c) Explain why ordinary polyalkenes, like poly(ethene), are electrical insulators. **[1]**

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10 Thermal Properties of Polymers: Thermoplastics vs Thermosets — 9701/41/M/J/19/Q10 [6 Marks] **HARD · Syllabus 35.3**

- (a) Explain the difference in structural bonding between a thermoplastic polymer and a thermosetting polymer. **[3]**

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- (b) Explain what happens when each type of polymer is heated, and relate this to their recycling potential. **[3]**

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11 Quantitative Analysis of Condensation Polymerisation: Degree of Polymerisation — 9701/42/O/N/21/Q10 [6 Marks] Syllabus 35.1

A sample of Nylon 6,6 has an average relative molecular mass (M_r) of 27,120.
Data: Repeat unit of Nylon 6,6 is $-\text{NH}-(\text{CH}_2)_6-\text{NH}-\text{CO}-(\text{CH}_2)_4-\text{CO}-$.

- (a) Calculate the relative formula mass of the repeat unit of Nylon 6,6. [2]

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- (b) Calculate the average degree of polymerisation, n , for this polymer sample. [2]

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- (c) Calculate the total mass of water eliminated during the synthesis of one mole of this polymer sample. [2]

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12 Comparison of Nylon 6 and Nylon 6,6 — 9701/42/M/J/18/Q10 [6 Marks] HARD · Syllabus 35.1

- (a) Nylon 6 is produced from a single monomer, caprolactam (or 6-aminohexanoic acid). Draw the repeat unit of Nylon 6. [2]

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- (b) Explain how the synthesis of Nylon 6 differs from that of Nylon 6,6 in terms of the number of different monomers used. [2]

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- (c) Compare the melting point and intermolecular forces in Nylon 6 with those in Nylon 6,6. [2]

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13 Biodegradability: Hydrolytic Cleavage of Poly(glycolic acid) (PGA) in Suture Threads — 9701/41/O/N/18/Q10 [6 Marks] HARD · Syllabus 35.4

Poly(glycolic acid), PGA, is a synthetic biodegradable polyester widely used in absorbable surgical sutures.
Monomer: Glycolic acid (hydroxyethanoic acid), $\text{HO}-\text{CH}_2-\text{COOH}$.

- (a) Draw the repeat unit of poly(glycolic acid). [2]

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- (b) Write an equation showing the hydrolysis of one ester bond in the polymer chain. [2]

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- (c) Explain why absorbable sutures made of PGA do not require surgical removal after a patient's wound has healed. [2]
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14 Polymer Recycling: Mechanical vs Chemical Feedstock Recycling — 9701/42/F/M/20/Q10 [6 Marks] HARD · Syllabus 35.4

- (a) Explain the principles of mechanical recycling of thermoplastic polymers and state one disadvantage. [2]
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- (b) Describe how PET (polyethylene terephthalate) bottles can undergo chemical recycling (feedstock recycling). [2]
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- (c) Explain why chemical recycling produces higher quality recycled polymers than mechanical recycling. [2]
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15 Polymer Linkages and Hydrogen Bonding Networks — 9701/41/M/J/17/Q9 [6 Marks] HARD · Syllabus 35.3

- (a) Draw a diagram showing two parallel chains of a polyamide with hydrogen bonds forming between them. [3]
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- (b) Explain why polyesters cannot form hydrogen bonds between adjacent chains unless hydroxy side-chains are present. [2]
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- (c) State how the difference in intermolecular bonding between polyamides and polyesters affects their melting temperatures. [1]
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16 Coordination Polymers and Metal–Organic Frameworks (MOFs) — 9701/42/O/N/17/Q10 [6 Marks] HARD · Syllabus 35.3

- (a) Explain what is meant by a metal-organic framework (MOF). [2]
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- (b) Identify the type of bonding between the metal ions and organic linkers in a MOF. [1]
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- (c) State two potential industrial applications of MOFs arising from their highly porous structures. [3]
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SECTION B: 4-MARK STRUCTURED EXAM QUESTIONS (40% TARIFF · Q17–Q32)

17 Repeat Unit and Monomers of Polycarbonate (Lexan) — 9701/42/M/J/23/Q11(b) [4 Marks] MEDIUM · Syllabus 35.1

- (a) Polycarbonate is synthesised from Bisphenol A and carbonyl dichloride (phosgene, COCl_2). Draw the repeat unit of polycarbonate. [2]
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- (b) Identify the small molecule eliminated and state the type of linkage formed. [2]
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18 Comparison of Monomer Functionality: Difunctional vs Monofunctional — 9701/41/O/N/22/Q9(b) [4 Marks] MEDIUM · Syllabus 35.1

- (a) Explain why condensation polymerisation requires monomers to have at least two functional groups per molecule. [2]
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- (b) Describe what happens if a small amount of a trifunctional monomer (e.g. propane-1,2,3-triol) is added during polyester synthesis. [2]
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19 Repeat Unit Deduction for Nomex Polyamide — 9701/42/M/J/22/Q10(b) [4 Marks] MEDIUM · Syllabus 35.3

- (a) Nomex is formed from benzene-1,3-diamine and benzene-1,3-dicarboxylic acid. Draw the repeat unit of Nomex. [2]
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- (b) Contrast the flexibility of Nomex with that of Kevlar (1,4-isomer) and relate this to fire-resistant suits. [2]
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20 Hydrolysis Products of Nylon 6,6 with Hot Dilute Hydrochloric Acid — 9701/41/M/J/21/Q11(b) [4 Marks] MEDIUM · Syllabus 35.4

- (a) Draw the structural formula of the nitrogen-containing product formed when Nylon 6,6 is completely hydrolysed by hot dilute HCl. [2]

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- (b) Draw the structural formula of the other organic product formed. [2]

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21 Comparing Atom Economy: Addition vs Condensation Polymerisation — 9701/42/O/N/20/Q9(b) [4 Marks] MEDIUM · Syllabus 35.1

- (a) Explain why the theoretical percentage atom economy for addition polymerisation is always 100%. [2]

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- (b) Explain why condensation polymerisation has an atom economy of less than 100%, and calculate the percentage atom economy for the synthesis of Nylon 6,6 from 1,6-diaminohexane and hexanedioic acid (Repeat unit Mr = 226, H₂O Mr = 18). [2]

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22 Biodegradation of Polyesters via Microbial Esterase Enzymes — 9701/43/M/J/23/Q10(b) [4 Marks] MEDIUM · Syllabus 35.4

- (a) State the role of bacterial esterase enzymes in the breakdown of biodegradable polyesters. [2]

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- (b) Explain why poly(ethene) is not attacked by bacterial esterase enzymes. [2]

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23 Deducing Monomer from Repeat Unit: Poly(3-hydroxybutanoic acid) (PHB) — 9701/42/F/M/22/Q11(b) [4 Marks] MEDIUM · Syllabus 35.2

PHB is a bacterial polyester with repeat unit: $-\text{O}-\text{CH}(\text{CH}_3)-\text{CH}_2-\text{CO}-$.

- (a) Give the systematic IUPAC name and displayed formula of the monomer from which PHB is formed. [2]

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- (b) State whether the monomer exhibits optical isomerism, giving a reason. [2]

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24 Chemical Recycling of PET via Methanolysis — 9701/41/O/N/23/Q10(b) [4 Marks]

MEDIUM · Syllabus 35.4

- (a) Write the word or chemical equation for the transesterification of PET with methanol. [2]

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- (b) State two advantages of this process over landfill disposal. [2]

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25 Properties of Thermosetting Epoxy Resins — 9701/42/M/J/20/Q9(b) [4 Marks]

MEDIUM · Syllabus 35.3

- (a) Explain the role of a 'hardener' (such as a diamine) in curing an epoxy resin. [2]

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- (b) Explain why cured epoxy resins are rigid, solvent-resistant, and do not melt upon heating. [2]

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26 Low-Density Poly(ethene) (LDPE) vs High-Density Poly(ethene) (HDPE) — 9701/41/M/J/19/Q10(b) [4 Marks]

MEDIUM · Syllabus 35.3

- (a) Contrast the chain branching in LDPE and HDPE. [2]

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- (b) Explain how the difference in chain structure affects the density and melting point of HDPE compared to LDPE. [2]

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27 Condensation of Hydroxy Acids: Polymerisation vs Lactide Formation — 9701/42/O/N/21/Q10(b) [4 Marks]

MEDIUM · Syllabus 35.1

- (a) When 2-hydroxyethanoic acid is heated, it can form a cyclic dilactone (glycolide). Draw the structure of glycolide. [2]

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- (b) Explain how ring-opening polymerisation of glycolide produces poly(glycolic acid) without eliminating water. [2]

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28 Comparison of Terylene and Nylon 6,6 Resistances to Acid and Alkali — 9701/42/M/J/18/Q10(b) [4 Marks] MEDIUM · Syllabus 35.4

- (a) Predict whether Terylene (polyester) or Nylon 6,6 (polyamide) is more susceptible to alkaline hydrolysis and explain why. [2]

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- (b) Predict whether Terylene or Nylon 6,6 is more susceptible to acid hydrolysis and explain why. [2]

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29 Polymers with Special Properties: Superabsorbent Polymers (Sodium Polyacrylate) — 9701/41/O/N/18/Q10(b) [4 Marks] MEDIUM · Syllabus 35.3

- (a) Draw the repeat unit of sodium polyacrylate, $-\text{[CH}_2\text{-CH(COONa)]}-$. [2]

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- (b) Explain in terms of osmosis and ion-dipole forces why sodium polyacrylate can absorb several hundred times its own mass of water. [2]

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30 Incineration of Polymers: Energy Recovery vs Toxic Emissions — 9701/42/F/M/20/Q10(b) [4 Marks] MEDIUM · Syllabus 35.4

- (a) State one major advantage of incinerating polymer waste compared to disposal in landfills. [1]

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- (b) Identify the toxic acidic gas produced when poly(chloroethene) (PVC) is incinerated, and explain how modern incinerators prevent its release. [3]

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31 Synthetic Rubber: Cis-trans Isomerism in Poly(isoprene) — 9701/41/M/J/17/Q9(b) [4 Marks] MEDIUM · Syllabus 35.3

- (a) Natural rubber is cis-1,4-poly(isoprene). Draw the structure of two repeat units of cis-poly(isoprene). [2]

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- (b) Explain how vulcanisation with sulfur alters the mechanical properties of natural rubber. [2]

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32 Deducing Monomers of Kevlar from Its Repeat Unit — 9701/42/O/N/17/Q10(b) [4 Marks]

MEDIUM · Syllabus 35.2

- (a) Draw the skeletal formula of benzene-1,4-diamine. [2]

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- (b) Draw the skeletal formula of benzene-1,4-dioic acid. [2]

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SECTION C: 2-MARK TARGETED EXAM QUESTIONS (20% TARIFF · Q33–Q40)

33 Classification of Nylon 6,6 and Terylene — 9701/42/M/J/23/Q11(a) [2 Marks]

EASY · Syllabus 35.1

- (a) Classify Nylon 6,6 and Terylene as a polyester or polyamide. [2]

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34 Small Molecule Eliminated in Polyesterification — 9701/41/O/N/22/Q9(a) [2 Marks]

EASY · Syllabus 35.1

- (a) Identify the small molecule eliminated when a diol reacts with a dicarboxylic acid, and when a diol reacts with a diacyl chloride. [2]

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35 Identifying Repeat Units — 9701/42/M/J/22/Q10(a) [2 Marks]

EASY · Syllabus 35.2

- (a) Define the term repeat unit of a polymer. [2]

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36 Intermolecular Forces in Polyamides — 9701/41/M/J/21/Q11(a) [2 Marks]

EASY · Syllabus 35.3

- (a) State the strongest type of intermolecular force present between adjacent polymer chains in Nylon 6,6. [2]

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37 Degradability of Polyalkenes vs Polyesters — 9701/42/O/N/20/Q9(a) [2 Marks]

EASY · Syllabus 35.4

- (a) State why polyesters can be hydrolysed by aqueous acid or alkali, whereas polyalkenes cannot. [2]

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38 Monomers of Terylene — 9701/43/M/J/23/Q10(a) [2 Marks]

EASY · Syllabus 35.1

- (a) Give the chemical formulas of the two monomers from which Terylene is manufactured. [2]

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39 Repeat Unit of Kevlar — 9701/42/F/M/22/Q11(a) [2 Marks]

EASY · Syllabus 35.3

- (a) Draw the skeletal formula of the repeat unit of Kevlar.

[2]

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40 Photodegradable Polymer Chromophores — 9701/41/O/N/23/Q10(a) [2 Marks]

EASY · Syllabus 35.4

- (a) Name the functional group commonly incorporated into polymer chains to make them photodegradable by absorbing UV radiation.

[2]

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SECTION D: HIGH-FREQUENCY CORE REPEATS — 10 MOST FREQUENTLY TESTED QUESTIONS (Q41–Q50)

41 Core Repeat 1: Deductions of Monomers from Condensation Polymers — 9701/42/M/J/22/Q9 [6 Marks]

HARD · Syllabus 35.2

The deconstruction of synthetic polymers into their constituent monomer units is one of the most frequently examined skills in Paper 4.

- (a) A polyamide has the repeat unit: $-\text{NH}-(\text{CH}_2)_4-\text{NH}-\text{CO}-(\text{CH}_2)_6-\text{CO}-$. Draw and name the two monomers.

[2]

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- (b) A polyester has the repeat unit: $-\text{O}-(\text{CH}_2)_3-\text{O}-\text{CO}-\text{C}_6\text{H}_4-\text{CO}-$. Draw and name the two monomers.

[2]

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- (c) Draw the repeat unit of the polymer formed when 4-aminobenzoic acid undergoes self-condensation, identifying the linkage type.

[2]

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42 Core Repeat 2: Chemical Degradability and Hydrolysis Mechanisms of Synthetic Polymers — 9701/41/O/N/21/Q10 [6 Marks]

HARD · Syllabus 35.4

- (a) Explain why poly(propene) cannot be broken down by boiling aqueous acid or alkali.

[2]

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- (b) Write the balanced chemical equation for the complete hydrolysis of Terylene by hot aqueous sodium hydroxide.

[2]

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- (c) Explain why alkaline hydrolysis is preferred over acid hydrolysis for recycling polyesters.

[2]

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43 Core Repeat 3: Kevlar Structure, Alignment, and High-Performance Properties — 9701/42/M/J/21/Q11 [6 Marks] HARD · Syllabus 35.3

- (a) State the systematic IUPAC names of the two monomers used to manufacture Kevlar. [2]
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- (b) Draw the repeat unit of Kevlar and indicate the points of hydrogen bonding with an adjacent chain. [2]
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- (c) Explain why Kevlar is five times stronger than steel on an equal weight basis. [2]
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44 Core Repeat 4: Polyesters from Diacyl Chlorides vs Dicarboxylic Acids — 9701/42/O/N/19/Q10 [6 Marks] HARD · Syllabus 35.1

A research chemist evaluates two synthetic routes to prepare a sample of Terylene:

Route 1: Benzene-1,4-dicarboxylic acid + Ethane-1,2-diol

Route 2: Benzene-1,4-dicarbonyl dichloride + Ethane-1,2-diol

- (a) State the reaction conditions required for Route 1, and name the byproduct. [2]
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- (b) State the reaction conditions required for Route 2, and name the byproduct. [2]
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- (c) Explain two advantages of Route 2 over Route 1 in a laboratory setting. [2]
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45 Core Repeat 5: Biodegradable Polymers: Structure and Hydrolysis of Polyglycolide — 9701/42/M/J/23/Q11(a) [4 Marks] HARD · Syllabus 35.4

- (a) Draw the repeat unit of poly(glycolide), which is formed from glycolic acid (hydroxyethanoic acid). [2]
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- (b) Explain how the ester linkages allow poly(glycolide) to break down harmlessly in the human body. [2]
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46 Core Repeat 6: Addition vs Condensation Polymerisation Comparison — 9701/41/O/N/22/Q9(c) [4 Marks] MEDIUM · Syllabus 35.1

- (a) Give two structural requirements for a monomer to undergo addition polymerisation. [2]

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- (b) Give two structural requirements for monomers to undergo condensation polymerisation. [2]

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47 Core Repeat 7: Repeat Unit of Kevlar vs Nylon 6,6 — 9701/42/F/M/21/Q11(b) [4 Marks] MEDIUM · Syllabus 35.3

- (a) Draw the repeat unit of Kevlar and state why its chains are much stiffer than Nylon 6,6. [2]

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- (b) Draw the repeat unit of Nylon 6,6 and explain why it is more flexible than Kevlar. [2]

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48 Core Repeat 8: Acid Hydrolysis of Polyamide Linkages — 9701/41/M/J/20/Q10(c) [4 Marks] MEDIUM · Syllabus 35.4

- (a) Write an equation showing the hydrolysis of an amide linkage ($-\text{CONH}-$) by aqueous acid ($\text{H}^+/\text{H}_2\text{O}$). [2]

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- (b) Explain why Nylon fabrics cannot be worn when handling concentrated sulfuric acid or hydrochloric acid. [2]

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49 Core Repeat 9: Linkage Type in Nylon 6,6 — 9701/42/M/J/23/Q11(e) [2 Marks] EASY · Syllabus 35.1

- (a) Name the functional linkage connecting monomer units in Nylon 6,6 and state which biological macromolecules contain the same linkage. [2]

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50 Core Repeat 10: Environmental Hazard of PVC Incineration — 9701/41/O/N/23/Q10(e) [2 Marks] EASY · Syllabus 35.4

- (a) Identify the corrosive acidic gas produced when poly(chloroethene) (PVC) is incinerated, and state one environmental harm it causes. [2]

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Mark Scheme — Topic 35 — Polymerisation

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — PAPER 4 MARKING CRITERIA

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
1(a)	Benzene-1,4-dicarboxylic acid (terephthalic acid) [1]; Ethane-1,2-diol (ethylene glycol) [1].	2
1(b)	$-\text{[NH}-(\text{CH}_2)_6\text{-NH-CO}-(\text{CH}_2)_4\text{-CO]}-$ [1]; Amide / peptide linkage ($-\text{CONH}-$) clearly identified [1].	2
1(c)	Addition polymerisation involves unsaturated monomers containing $\text{C}=\text{C}$ double bonds with no loss of atoms (100% atom economy) [1]; Condensation polymerisation involves bifunctional or polyfunctional monomers reacting with the elimination of small molecules such as H_2O or HCl [1].	2
2(a)	Polyalkenes possess a continuous saturated backbone of non-polar $\text{C}-\text{C}$ and $\text{C}-\text{H}$ bonds [1]; They contain no polar sites to attract electrophiles, nucleophiles, or microbial enzymes, resisting chemical attack and enzymatic cleavage [1].	2
2(b)	Polyesters and polyamides contain polar carbonyl groups ($\text{C}=\text{O}$) in their ester and amide linkages [1]; The electron-deficient carbonyl carbon is vulnerable to nucleophilic attack by water or hydroxide ions (hydrolysis), breaking the polymer into monomer units [1].	2
2(c)	The amide bonds undergo alkaline hydrolysis (saponification) [1]; The polymer is cleaved into 1,6-diaminohexane and the sodium salt of hexanedioic acid (disodium hexanedioate) [1].	2
3(a)	$-\text{[NH-C}_6\text{H}_4\text{-NH-CO-C}_6\text{H}_4\text{-CO]}-$ showing 1,4-disubstituted benzene rings linked by amide groups [2].	2
3(b)	Benzene-1,4-diamine (1,4-diaminobenzene) [1]; Benzene-1,4-dicarbonyl dichloride (or benzene-1,4-dicarboxylic acid) [1].	2
3(c)	The rigid 1,4-phenylene rings maintain straight, linear, planar polymer chains that pack closely together [1]; Highly ordered and dense arrays of intermolecular hydrogen bonds between $\text{C}=\text{O}$ and $\text{N}-\text{H}$ groups of adjacent chains, reinforced by π - π stacking, resist slippage and tensile deformation [1].	2
4(a)	Ester linkage ($-\text{COO}-$) [1]; Amide / peptide linkage ($-\text{CONH}-$) [1].	2
4(b)	Monomer 1: Butanedioic acid, $\text{HOOC-CH}_2\text{CH}_2\text{-COOH}$ [1]; Monomer 2: Propane-1,2-diol, $\text{HO-CH(CH}_3\text{)CH}_2\text{-OH}$ [1]; Monomer 3: Benzene-1,4-dicarboxylic acid, $\text{HOOC-C}_6\text{H}_4\text{-COOH}$ [1]; Monomer 4: Butane-1,4-diamine, $\text{H}_2\text{N}-(\text{CH}_2)_4\text{-NH}_2$ [1].	4
5(a)	$-\text{[O-CH(CH}_3\text{)-CO]}-$ [2].	2
5(b)	Lactic acid is bifunctional, containing both a nucleophilic hydroxy group ($-\text{OH}$) and an electrophilic carboxylic acid group ($-\text{COOH}$) within the same molecule [1]; Ethanoic acid has only one functional group ($-\text{COOH}$) and cannot form continuous chains [1].	2
5(c)	PLA is derived from renewable biological resources (e.g. fermented corn starch / sugarcane) rather than finite petrochemicals [1]; PLA is biodegradable and compostable, breaking down into non-toxic lactic acid / CO_2 / H_2O via bacterial enzymatic hydrolysis [1].	2
6(a)	The two solvent layers (aqueous and organic) are immiscible [1]; The monomers can only come into contact and react where the two phases meet at the boundary interface [1].	2
6(b)	The reaction eliminates toxic, acidic hydrogen chloride (HCl) gas [1]; NaOH neutralises the HCl to form NaCl and water, preventing protonation of the amine which would deactivate it as a nucleophile [1].	2
6(c)	Acyl chlorides are far more reactive than carboxylic acids and react rapidly at room temperature [1]; Dicarboxylic acids require elevated temperatures and strong acid catalysis, which would not work in a cold beaker demonstration [1].	2
7(a)	The carbonyl group acts as a chromophore that absorbs solar ultraviolet (UV) radiation [1]; Absorption of UV energy promotes electrons to antibonding orbitals, causing homolytic cleavage of carbon-carbon bonds adjacent to the carbonyl (Norrish type I/II reactions) [1]; This fragments the long polymer chains into short oligomers that can subsequently be consumed by micro-organisms [1].	3
7(b)	Repeat unit: $-\text{[CH}_2\text{-CH(OH)-]}-$ [1]; Every monomer unit contains a polar hydroxy group ($-\text{OH}$) [1]; These $-\text{OH}$ groups form extensive hydrogen bonds with surrounding water molecules, releasing sufficient hydration energy to overcome polymer-polymer interactions and dissolve the chain [1].	3
8(a)	Condensation polymerisation [1]; Water, H_2O [1].	2
8(b)	Glycosidic bond (or ether linkage / $-\text{C-O-C}-$) [1].	1

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
8(c)	Starch hydrolysis is catalysed by biological enzymes (e.g. amylase) under mild conditions (pH ~7, 37 °C) with high stereospecificity [1]; Industrial hydrolysis of polyesters requires harsh conditions such as concentrated aqueous acid or alkali under prolonged reflux at high temperatures [1]; Both break down the polymer backbone into monomer units via addition of water across the functional linkage [1].	3
9(a)	$-\text{[CH=CH-CH=CH-CH=CH]}-$ with trans alternating stereochemistry [2].	2
9(b)	The p-orbitals on adjacent sp ² carbon atoms overlap sideways continuously along the entire polymer backbone [1]; This creates an extended, delocalised π -electron system [1]; When doped with an oxidising or reducing agent (e.g. I ₂), electrons or holes are introduced into the delocalised band and can move freely along the chain under an applied potential difference [1].	3
9(c)	In poly(ethene), all carbon atoms are sp ³ hybridised with all valence electrons tightly localised in σ -bonds, so there are no mobile delocalised electrons to carry charge [1].	1
10(a)	Thermoplastics consist of discrete linear or branched polymer chains held together only by intermolecular forces (London forces, dipole-dipole, or hydrogen bonds) [1]; Thermosets contain permanent covalent cross-links linking adjacent chains into a vast three-dimensional network [2].	3
10(b)	On heating, thermoplastics soften and melt because weak intermolecular forces are easily overcome, allowing them to be reshaped and recycled repeatedly [1]; Thermosets do not melt on heating because rigid covalent cross-links cannot break without decomposing (charring) the polymer [1]; Thus, thermosets cannot be remoulded or recycled by melting [1].	3
11(a)	$\text{C}_{12}\text{H}_{22}\text{N}_2\text{O}_2: (12 \times 12.0) + (22 \times 1.0) + (2 \times 14.0) + (2 \times 16.0) = 144 + 22 + 28 + 32 = 226.0$ [2].	2
11(b)	Degree of polymerisation $n = M_r / M_r(\text{repeat unit}) = 27,120 / 226.0 = 120$ [2].	2
11(c)	Each repeat unit formed eliminates 2 molecules of H ₂ O [1]; Total moles of H ₂ O eliminated per mole of polymer = $2n = 240$ mol; Mass of H ₂ O = $240 \times 18.0 = 4320$ g [1].	2
12(a)	$-\text{[NH-(CH}_2)_5\text{-CO]}-$ [2].	2
12(b)	Nylon 6 is made from a single monomer that contains both an amino group and a carboxyl group (or via ring opening of caprolactam) [1]; Nylon 6,6 is a copolymer made from two different bifunctional monomers (a diamine and a dicarboxylic acid) [1].	2
12(c)	Both possess identical amide ($-\text{CONH}-$) linkages and exhibit intermolecular hydrogen bonding [1]; Nylon 6,6 has a slightly higher melting point (~ 265 °C vs ~ 220 °C) because its symmetrical structure allows more regular and tighter hydrogen-bonded packing [1].	2
13(a)	$-\text{[O-CH}_2\text{-CO]}-$ [2].	2
13(b)	$-\text{O-CH}_2\text{-CO-} + \text{H}_2\text{O} \rightarrow -\text{O-CH}_2\text{-COOH} + \text{HO-}$ (or ester + H ₂ O $\rightarrow$ $-\text{OH} + -\text{COOH}$) [2].	2
13(c)	Body fluids contain water and esterase enzymes that slowly hydrolyse the ester bonds over several weeks [1]; The breakdown product (glycolic acid) is naturally metabolised by normal cellular respiration into CO ₂ and H ₂ O and safely excreted [1].	2
14(a)	Polymers are sorted, washed, shredded, melted, and remoulded into new products [1]; Repeated heating cycles cause thermal degradation / chain shortening, reducing physical strength [1].	2
14(b)	PET is depolymerised by heating with methanol or ethylene glycol (methanolysis / glycolysis) or aqueous alkali [1]; This breaks the ester bonds and recovers pure monomer building blocks (dimethyl terephthalate and ethane-1,2-diol) [1].	2
14(c)	The recovered monomers can be separated, purified, and repolymerised into virgin-quality polymer [1]; Any dyes, additives, or contaminants are completely removed during monomer purification [1].	2
15(a)	Two chains showing C=O and N-H groups aligned [1]; Dotted line showing hydrogen bond between carbonyl oxygen and N-H hydrogen [1]; Correct partial charges (δ^- on O, δ^+ on H) and lone pair shown on oxygen [1].	3
15(b)	Polyesters contain carbonyl oxygen atoms (hydrogen bond acceptors) but lack hydrogen atoms bonded to highly electronegative atoms (no N-H or O-H hydrogen bond donors) [2].	2
15(c)	Polyamides have significantly higher melting temperatures than polyesters of similar molecular weight due to strong intermolecular hydrogen bonding [1].	1

Q	EXAMINER MARKING POINTS / WORKED MATHEMATICAL DERIVATIONS	MARKS
16(a)	A coordination network consisting of metal ions or clusters coordinated to polydentate organic bridging ligands (linkers) forming a 1D, 2D, or 3D porous structure [2].	2
16(b)	Coordinate (dative covalent) bonding [1].	1
16(c)	Gas storage (e.g. hydrogen fuel or methane storage in vehicles) [1]; Carbon dioxide capture and sequestration [1]; Heterogeneous catalysis or drug delivery [1].	3
17(a)	$-\text{[O-C}_6\text{H}_4\text{-C(CH}_3\text{)}_2\text{-C}_6\text{H}_4\text{-O-CO]-}$ [2].	2
17(b)	Hydrogen chloride, HCl [1]; Carbonate ester linkage ($-\text{O-CO-O-}$) [1].	1
18(a)	Each reacting end must leave a free functional group available to react with another monomer, enabling continuous chain growth at both ends [2].	2
18(b)	The third functional group acts as a branch point [1]; This leads to cross-linking between separate polymer chains, creating a rigid thermosetting 3D network [1].	2
19(a)	$-\text{[NH-(1,3-C}_6\text{H}_4\text{)-NH-CO-(1,3-C}_6\text{H}_4\text{)-CO]-}$ [2].	2
19(b)	The 1,3-linkages introduce bends / kinks into the polymer chain, making Nomex more flexible and easier to weave into textiles than rigid Kevlar [1]; Nomex retains excellent thermal resistance and does not melt or drip when exposed to flames [1].	2
20(a)	Hexane-1,6-diammonium dichloride, $+\text{H}_3\text{N-(CH}_2\text{)}_6\text{-NH}_3^+ 2\text{Cl}^-$ [2].	2
20(b)	Hexanedioic acid, $\text{HOOC-(CH}_2\text{)}_4\text{-COOH}$ [2].	2
21(a)	All atoms from the monomer molecules are incorporated into the polymer chain; no waste byproducts are formed [2].	2
21(b)	Small molecules (such as H_2O) are eliminated as waste [1]; Total mass of reactants = $226 + (2 \times 18) = 262$; % atom economy = $(226 / 262) \times 100 = 86.3\%$ [1].	2
22(a)	Esterases bind to the polar ester linkages ($-\text{COO-}$) in the active site and catalyse the addition of water, cleaving the ester bonds into non-toxic carboxylic acids and alcohols [2].	2
22(b)	Poly(ethene) contains only non-polar C-C bonds with no ester linkages; its hydrophobic backbone cannot fit into the active site of esterases [2].	2
23(a)	3-hydroxybutanoic acid [1]; $\text{CH}_3\text{-CH(OH)-CH}_2\text{-COOH}$ with all bonds displayed [1].	2
23(b)	Yes [1]; Carbon-3 is a chiral centre bonded to four different groups ($-\text{H}$, $-\text{CH}_3$, $-\text{OH}$, $-\text{CH}_2\text{COOH}$) [1].	2
24(a)	PET + methanol dimethyl terephthalate + ethane-1,2-diol [2].	2
24(b)	Conserves fossil petrochemical resources by recycling monomers indefinitely [1]; Reduces waste accumulation in landfills and prevents environmental microplastic pollution [1].	2
25(a)	The diamine hardener contains reactive $-\text{NH}_2$ groups that open epoxide rings on separate resin molecules [1]; This establishes covalent cross-links between the polymer chains [1].	2
25(b)	The vast 3D network of covalent cross-links prevents chains from sliding past one another (giving high rigidity) [1]; Solvents cannot separate covalently bound chains, and thermal energy causes decomposition before melting can occur [1].	2
26(a)	LDPE has significant chain branching [1]; HDPE consists of linear, unbranched polymer chains [1].	2
26(b)	Linear HDPE chains can pack much closer and more regularly together into crystalline regions [1]; This increases London dispersion forces between chains, resulting in higher density and a higher melting point than LDPE [1].	2
27(a)	Six-membered cyclic ring with two ester groups ($-\text{O-CO-}$) formed by condensation of two molecules of 2-hydroxyethanoic acid [2].	2
27(b)	A catalyst opens the cyclic ester ring, generating an active chain end that attacks another cyclic monomer [1]; Because all atoms of the cyclic monomer are incorporated into the chain, it behaves as an addition-type ring-opening polymerisation with no small molecule eliminated [1].	2

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28(a)	Terylene is more susceptible to alkali [1]; The ester carbonyl carbon is more electrophilic than an amide carbonyl (due to greater electronegativity of oxygen over nitrogen) and is attacked rapidly by OH ⁻ [1].	2
28(b)	Nylon 6,6 is more susceptible to strong acid [1]; The nitrogen/carbonyl of the amide linkage is readily protonated by H ⁺ , greatly activating the carbonyl carbon towards attack by water [1].	2
29(a)	-[CH ₂ -CH(COO ⁻ Na ⁺)]- showing carboxylate sodium salt on the chain [2].	2
29(b)	The ionic -COO ⁻ and Na ⁺ groups are strongly hydrated by water molecules through ion-dipole interactions [1]; The high concentration of ions inside the cross-linked polymer network draws water into the polymer by osmosis, causing it to swell dramatically into a gel without dissolving [1].	2
30(a)	Massive reduction in waste volume and recovery of thermal energy for electricity generation / district heating [1].	1
30(b)	Hydrogen chloride, HCl [1]; Flue gases are passed through wet scrubbers containing a basic substance such as calcium oxide (CaO) or calcium carbonate (CaCO ₃) [1]; The basic substance neutralises the acidic HCl to produce harmless solid calcium chloride: CaO + 2HCl → CaCl ₂ + H ₂ O [1].	3
31(a)	Correct cis-configuration around the C=C double bonds in adjacent isoprene repeat units (-[CH ₂ -C(CH ₃)=CH-CH ₂]-) [2].	2
31(b)	Sulfur atoms form covalent disulfide cross-links (-S-S-) between adjacent polyisoprene chains [1]; These cross-links prevent chains from sliding permanently past one another when stretched, greatly improving elasticity, resilience, and hardness [1].	2
32(a)	Benzene ring with two amino groups (-NH ₂) at positions 1 and 4 [2].	2
32(b)	Benzene ring with two carboxyl groups (-COOH) at positions 1 and 4 [2].	2
33(a)	Nylon 6,6 is a polyamide [1]; Terylene is a polyester [1].	2
34(a)	Water, H ₂ O (with dicarboxylic acid) [1]; Hydrogen chloride, HCl (with diacyl chloride) [1].	2
35(a)	The specific arrangement of atoms or repeating block derived from the monomers [1]; which is repeated successively to generate the complete polymer macromolecule [1].	2
36(a)	Hydrogen bonding [1]; Between the C=O oxygen of one chain and the N-H hydrogen of an adjacent chain [1].	2
37(a)	Polyesters contain polar C=O and C-O bonds susceptible to nucleophilic attack [1]; Polyalkenes contain only non-polar C-C and C-H bonds [1].	2
38(a)	HOOC-C ₆ H ₄ -COOH [1]; HO-CH ₂ CH ₂ -OH [1].	2
39(a)	-[NH-(1,4-C ₆ H ₄)-NH-CO-(1,4-C ₆ H ₄)-CO]- [2].	2
40(a)	Carbonyl group / C=O (or ketone / aldehyde chromophore) [2].	2
41(a)	Butane-1,4-diamine (1,4-diaminobutane), H ₂ N-(CH ₂) ₄ -NH ₂ [1]; Octanedioic acid, HOOC-(CH ₂) ₆ -COOH [1].	2
41(b)	Propane-1,3-diol, HO-(CH ₂) ₃ -OH [1]; Benzene-1,4-dicarboxylic acid, HOOC-C ₆ H ₄ -COOH [1].	2
41(c)	-[NH-C ₆ H ₄ -CO]- [1]; Aromatic polyamide / amide linkage [1].	2
42(a)	Poly(propene) has a saturated non-polar carbon-carbon backbone with no delta positive atoms to attract nucleophilic attack by H ₂ O or OH ⁻ [2].	2
42(b)	-[CO-C ₆ H ₄ -CO-O-CH ₂ CH ₂ -O] _n - + 2n NaOH → n Na ⁺ -OOC-C ₆ H ₄ -COO-Na ⁺ + n HOCH ₂ CH ₂ OH [2].	2
42(c)	Alkaline hydrolysis produces carboxylate salts that are water-soluble and cannot re-esterify, making the reaction completely irreversible and quantitative [1]; Acid hydrolysis is a reversible equilibrium reaction that does not go to completion [1].	2
43(a)	Benzene-1,4-diamine (or 1,4-diaminobenzene) [1]; Benzene-1,4-dicarbonyl dichloride (or terephthaloyl chloride) [1].	2
43(b)	-[NH-C ₆ H ₄ -NH-CO-C ₆ H ₄ -CO]- [1]; Dotted line showing hydrogen bonding between N-H and C=O of adjacent chains [1].	2

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43(c)	Linear, uninked chains allow maximum surface contact and regular arrays of strong intermolecular hydrogen bonds [1]; Planar aromatic rings allow π - π ; orbital stacking, and low density gives an extremely high strength-to-weight ratio [1].	2
44(a)	Concentrated acid catalyst (e.g. H ₂ SO ₄) and high temperature / heating under reflux [1]; Water, H ₂ O [1].	2
44(b)	Room temperature (no catalyst required) [1]; Hydrogen chloride gas, HCl [1].	2
44(c)	Acyl chlorides are vastly more reactive than carboxylic acids, giving high yields rapidly at room temperature [1]; The reaction is irreversible because HCl is eliminated as a gas, whereas Route 1 is an equilibrium reaction [1].	2
45(a)	-[O-CH ₂ -CO]- [2].	2
45(b)	Ester linkages undergo slow hydrolytic cleavage by water in body tissues, breaking the polymer into glycolic acid [1]; Glycolic acid is a non-toxic natural metabolite that is broken down into carbon dioxide and water [1].	2
46(a)	Must contain a carbon-carbon double bond (C=C) / unsaturation [1]; Polymerisation occurs by breaking the π -bond without loss of atoms [1].	2
46(b)	Must have at least two functional groups per molecule (bifunctional or polyfunctional) [1]; Must possess reactive groups such as -OH, -COOH, -NH ₂ , or -COCl that can react together with the elimination of a small molecule [1].	2
47(a)	-[NH-C ₆ H ₄ -NH-CO-C ₆ H ₄ -CO]- [1]; The rigid planar 1,4-phenylene rings prevent chain twisting and rotation, keeping chains linear and stiff [1].	2
47(b)	-[NH-(CH ₂) ₆ -NH-CO-(CH ₂) ₄ -CO]- [1]; Contains flexible aliphatic -(CH ₂)- methylene chains that can freely rotate around C-C σ -bonds [1].	2
48(a)	-CO-NH- + H ₂ O + H ⁺ -COOH + -NH ₃ ⁺ [2].	2
48(b)	Concentrated acids rapidly protonate and hydrolyse the amide bonds along the polymer chains [1]; This breaks the macromolecules into short fragments, destroying the fabric's tensile strength and creating holes [1].	2
49(a)	Amide linkage / peptide bond (-CONH-) [1]; Proteins / polypeptides [1].	2
50(a)	Hydrogen chloride, HCl [1]; Causes acid rain which damages aquatic ecosystems / dissolves limestone buildings / causes respiratory distress [1].	2